

Experimental Optimisation 1: Baud Rate

Group 6

May 2026

Background

The focus of this project is improving communication reliability in a backscatter system, evaluated using Bit Error Rate (BER), effective data rate, and experiment duration.

In a backscatter system, the tag does not transmit actively. Instead, it reflects and modulates the carrier signal generated by a separate carrier node. The reflected signal is received by a receiver node. The overall path loss depends on both the carrier-to-tag link and the tag-to-receiver link, making the system particularly sensitive to tag position.

The baud rate determines the number of symbols transmitted per second. Each symbol occupies a time window of duration $T_s = 1/f_{\text{baud}}$. The receiver must correctly sample the incoming signal within this window to decode each bit. At high baud rates, T_s is short, leaving little margin for correct detection under weak or noisy conditions. Lowering the baud rate increases T_s , giving the receiver more time to correctly detect the incoming signal and reducing the probability of a bit error.

However, an optimal baud rate exists. Below this point, further reduction may introduce additional timing or synchronisation issues and no longer improves performance.

A sync word is used to ensure that only intended packets are accepted by the receiver. However, the sync word only gates packet acceptance — it does not protect the payload. Once a packet passes the sync word check, the payload can still contain bit errors. BER within received packets therefore remains a meaningful metric for evaluating communication reliability.

Hypothesis

Reducing the baud rate from the baseline of 100k will improve communication reliability by increasing T_s and giving the receiver more time to correctly detect each transmitted symbol. An optimal baud rate exists beyond which further reduction no longer improves performance. Configurations that reduce BER at the measured edge positions may not necessarily improve communication at the midpoint position, where path loss is highest.

Experimental Setup

The experimental setup consisted of a carrier generator and a receiver node separated by approximately 3 m. The tag was placed at five positions along the line between the carrier generator and receiver.

The tag positions, defined as distance from carrier generator / distance to receiver, were:

- P1: 27 cm / 273 cm
- P2: 50 cm / 250 cm
- P3: 150 cm / 150 cm (midpoint)
- P4: 250 cm / 50 cm
- P5: 277 cm / 23 cm

The midpoint position P3 represents the worst-case scenario for backscatter communication, as both links experience equal and maximum path loss.

Packets were continuously transmitted by the tag. The receiver logged incoming packets until 200 packets were successfully received. The experiment duration Δt was computed as the timestamp difference between the first and last received packet:

$$\Delta t = t_{last} - t_{first}$$

For the midpoint position P3, measurements were continued for up to 5 minutes. If no packets were received within this period, the configuration was recorded as a communication failure with $BER = 100\%$ and $D/R = 0$ byte/s.

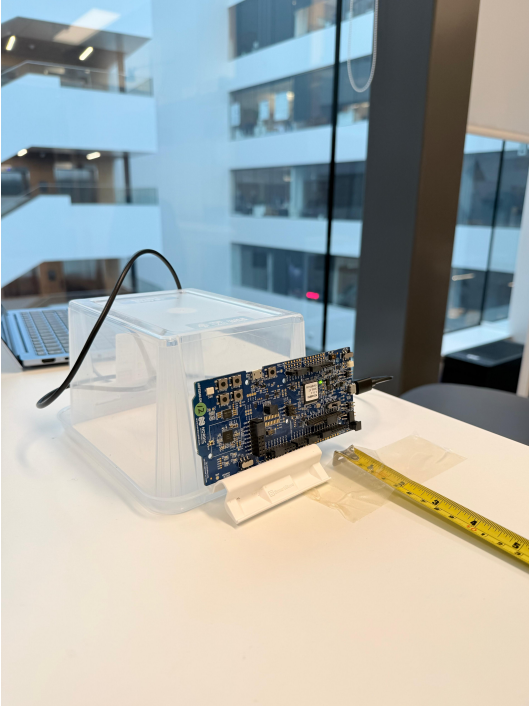


Figure 1: The nRF52840 carrier generator node mounted vertically in a plastic enclosure.

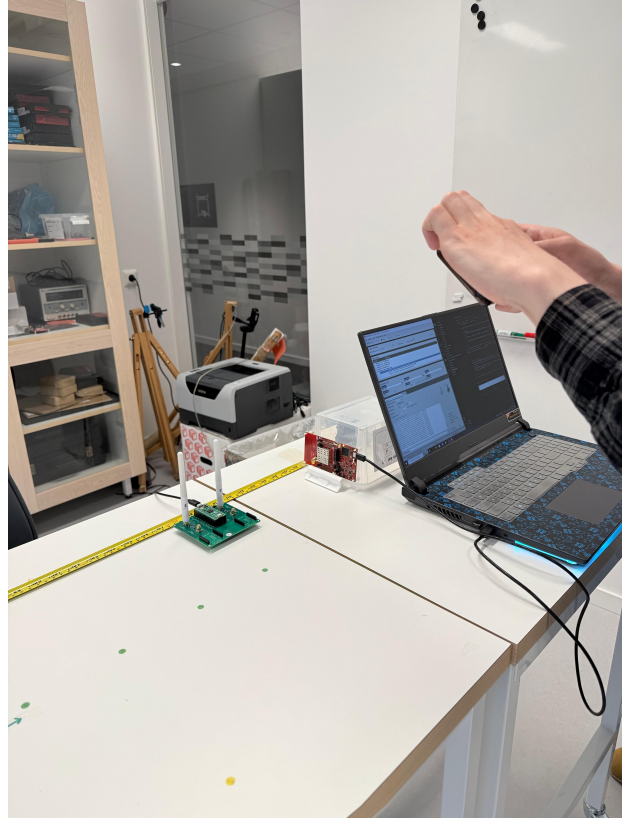


Figure 2: Experimental setup showing the tag (left, green board with antennas) and the CC1352 receiver node (right, red board) connected to a laptop running SmartRF. A measuring tape marks the tag positions.

Metrics

Three metrics were used to evaluate performance.

Bit Error Rate (BER) is the ratio of incorrectly received bits to the total number of received bits:

$$BER = \frac{\text{Number of wrong bits}}{\text{Total number of received bits}}$$

BER is computed only over received packets and does not account for lost packets.

Effective data rate is defined as:

$$D/R = \frac{N_{rx} \times L_{payload}}{\Delta t}$$

where N_{rx} is the total number of packets successfully received and logged by the receiver, $L_{payload} = 12$ bytes is the payload size per packet (excluding the 2-byte pseudo-sequence number field), and Δt is the experiment duration in seconds. The unit is byte/s. Although the given statistics script labels this metric as bit/s, the formula computes $N_{rx} \times 12 \text{ bytes}/\Delta t$, which gives byte/s. This is confirmed by the measured values — for example, baseline P1 gives $205 \times 12/84.42 \approx 29.1$ byte/s, consistent with the recorded value of 29.16. This metric reflects packet loss — missing packets reduce N_{rx} and therefore reduce the effective data rate.

Experiment duration Δt is the time from the first to the last received packet in seconds.

These three metrics are used together because BER alone does not capture the full picture. A configuration with low BER but high packet loss can still result in poor throughput and long experiment duration.

Method

The baud rate was varied in the firmware while keeping the physical setup unchanged. Measurements were performed for baud rates of 100k (baseline), 80k, 70k, 60k, and 50k.

The baseline configuration at 100k baud was repeated three times in a controlled environment and the results were averaged to reduce the effect of environmental variability. All other configurations were measured once. The physical setup and tag positions were kept unchanged across all measurements.

Results

Table 1 shows all individual measurements. For configurations where no packets were received at P3 within 5 minutes, BER is recorded as 100% and D/R as 0 byte/s.

Table 1: BER, effective data rate, and experiment duration for each baud rate configuration and tag position. Baseline values are averaged over three trials.

Baud Rate	Position	BER (%)	D/R (byte/s)	Time (s)
100k (baseline)	P1	10.66	29.16	84.42
	P2	13.03	26.95	89.52
	P3	100.00	0.00	—
	P4	18.51	20.99	116.32
	P5	15.33	24.55	97.89
80k	P1	4.30	37.03	64.81
	P2	10.00	31.71	75.68
	P3	100.00	0.00	—
	P4	8.16	34.03	70.52
	P5	6.95	32.15	74.64
70k	P1	3.72	39.72	60.66
	P2	9.70	30.87	77.75
	P3	29.96	11.73	204.53
	P4	15.00	23.49	102.14
	P5	11.94	29.48	81.39
60k	P1	3.57	37.03	64.79
	P2	8.64	33.66	71.28
	P3	100.00	0.00	—
	P4	10.97	26.68	89.44
	P5	8.25	32.48	73.88
50k	P1	11.50	25.64	93.58
	P2	14.40	17.17	139.72
	P3	100.00	0.00	—
	P4	18.45	14.44	166.16
	P5	14.69	22.02	108.09

Table 2 shows the averages computed over all five positions per configuration, including communication failures at P3.

Table 2: Average BER, effective data rate, and experiment duration per baud rate configuration. BER and D/R averages include P3 failures (BER = 100%, D/R = 0). Time averages are computed only over positions where packets were received.

Baud Rate	Avg BER (%)	Avg D/R (byte/s)	Avg Time (s)
100k (baseline)	31.51	20.33	97.04
80k	25.88	26.98	71.41
70k	14.06	27.06	105.30
60k	26.29	25.97	74.85
50k	31.81	15.85	126.89

Figure 3 shows the baseline BER per position. Figure 4 shows the BER comparison across all configurations and positions. Figure 5 shows the average BER, effective data rate, and experiment duration per configuration.

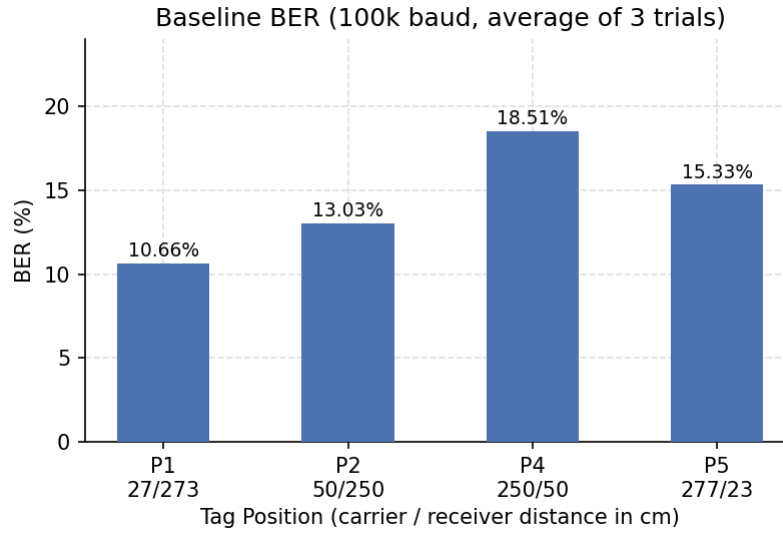


Figure 3: Average BER per tag position for the baseline 100k configuration, averaged over three trials. P3 is excluded as no packets were received within 5 minutes.

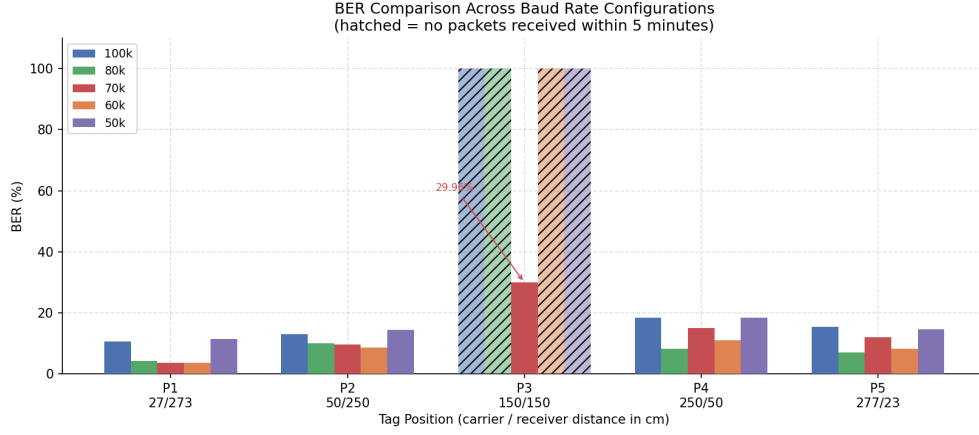


Figure 4: BER per tag position for all baud rate configurations. The 70k configuration is the only one where packets were received at the midpoint position P3. Hatched bars indicate positions where no packets were received within 5 minutes, recorded as BER = 100%.

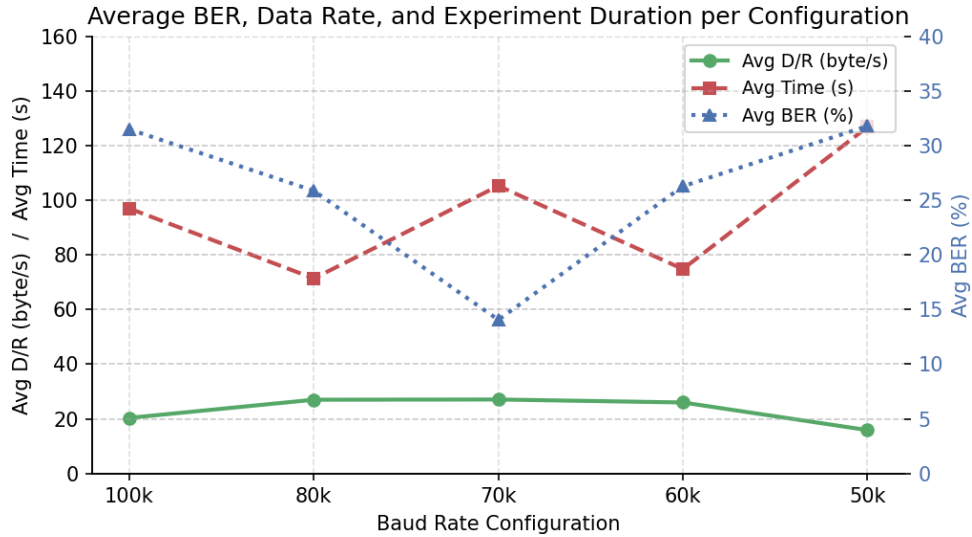


Figure 5: Average BER (right axis, %), effective data rate D/R (left axis, byte/s), and experiment duration (left axis, s) per baud rate configuration. BER and D/R averages include P3 failures. Time is averaged only over positions where packets were received. D/R and experiment duration are plotted on the same left axis for trend comparison; they share the axis scale but have different units.

Discussion and Conclusion

The results confirm the hypothesis that reducing the baud rate improves reliability up to an optimal point.

80k baud — best performance at edge positions. At 80k baud, the longer bit duration gave the receiver more time to correctly detect the incoming signal, resulting in the lowest BER and highest effective data rate among the edge positions. However, the 80k configuration did not receive any packets at the midpoint position P3 within 5 minutes.

70k baud — only configuration to reach the midpoint. Only the 70k configuration successfully received packets at P3. The longer bit duration at 70k increased T_s sufficiently for the receiver to detect some packets at the midpoint, where both the carrier-to-tag and tag-to-receiver links experience equal and maximum path loss. However, the BER at this position was approximately 30% and the experiment duration exceeded 200s, indicating that the link was marginal. This demonstrates that successful packet reception alone does not imply reliable communication.

Overall best configuration. When the midpoint failures are included in the average, 70k achieves the best overall BER of 14.06%, the highest average effective data rate of 27.06 byte/s, and the shortest average experiment duration of 105.30s (averaged over all five positions including P3). It is important to note that this is partly because 70k is the only configuration with an actual measurement at P3 (BER = 29.96%), while all other configurations contribute BER = 100% and D/R = 0 byte/s at P3. This significantly penalises the averages for 100k, 80k, 60k, and 50k.

60k and 50k — no improvement. At 60k and 50k, further reduction did not improve performance. BER increased again and the effective data rate decreased, consistent with the hypothesis that an optimal baud rate exists beyond which further reduction no longer improves performance.

Conclusion. The results demonstrate that communication system optimisation involves a trade-off between reliability at edge positions and coverage at challenging positions. A configuration that minimises BER at edge positions does not necessarily perform best across the full measurement range. The optimal baud rate depends on the system requirements: 80k is preferred when reliability at reachable positions is the priority, while 70k provides better overall coverage.